INCH-POUND
MIL-PRF-3098/122E
28 July 2011
SUPERSEDING
MIL-PRF-3098/122D
27 July 2005

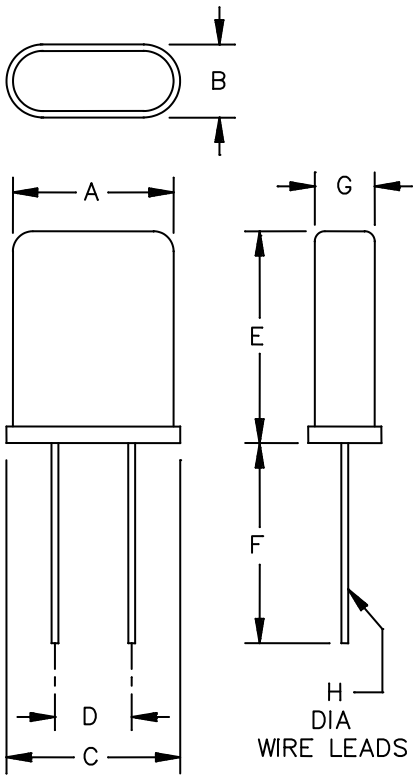
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR141/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 48 MHz to 90 MHz; fifth mechanical overtone; noncontrolled; series resonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.016	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR141/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 48.0 MHz to 90 MHz, inclusive.

Frequency tolerance:

Primary operating temperature range: ±25 ppm.

Secondary operating temperature range: ±45 ppm.

Equivalent resistance: 50 ohms.

Mode of oscillator: Fifth mechanical overtone.

Operating temperature ranges:

Primary: -30°C to ±55°C, inclusive.

Secondary: -55°C to -30°C and +55°C to +65°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF, maximum.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 204, MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR141/U or CR141B/U (followed by specified frequency). For Product Level S, PIN shall be CR141S/U (followed by specified frequency, see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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(Project 5955-2011-009)

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